TP 07: Multi-Armed Bandits for Online Ads

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1 Introduction

The multi-armed bandit problem is a fundamental challenge in online systems. We have multiple ads (arms) with unknown performance. At each round, we must choose which ad to show. This creates a trade-off: should we explore new ads to learn their performance, or exploit the ads we already know work best?

This lab compares three algorithms for solving this problem on a real advertising dataset with 10 ads over 10,000 rounds.

2 Methods

2.1 Upper Confidence Bound (UCB)

UCB assumes each ad has an unknown success rate. At each round, it calculates an optimistic estimate for each ad:

$$\text{UCB}_i = \bar{r}_i + \sqrt{\frac{\ln(n)}{N_i}}$$

where $\bar{r}_i$ is the average reward, N_i is times selected, and n is the current round. It selects the ad with the highest upper bound. This balances exploration and exploitation automatically.

2.2 Thompson Sampling

Thompson Sampling maintains a probability distribution over each ad's success rate. For binary rewards, it uses Beta distributions. At each round, it samples from each ad's distribution and selects the one with the highest sample. Uncertainty naturally decreases with experience.

2.3 ϵ-Greedy

ϵ-Greedy uses a fixed exploration rate. With probability ϵ, it explores randomly. With probability $1-\epsilon$, it exploits by selecting the ad with the highest observed average reward.

3 Experiments

3.1 Setup

Dataset: 10 ads, 10,000 rounds. We measured:

- Total cumulative reward (clicks)
- Optimal ad selection rate (%)
- Cumulative regret

ϵ-Greedy tested with: $\epsilon \in \{0.01, 0.05, 0.1, 0.2, 0.3\}$

3.2 Results Table

Algorithm	Total Reward	Optimal %	Regret
Thompson Sampling	2,613	94.86%	82
ϵ-Greedy (0.1)	2,551	90.57%	144
ϵ-Greedy (0.05)	2,592	92.15%	103
UCB	2,178	63.23%	517
ϵ-Greedy (0.5)	1,935	53.44%	760

Table 1: Algorithm Performance

4 Results & Discussion

4.1 Algorithm Comparison

Thompson Sampling achieved the highest reward (2,613 clicks, 94.86% optimal selection). It also has the lowest regret (82).

UCB obtained 2,178 clicks (20% worse than Thompson). It converges slower, selecting the optimal ad only 63% of the time.

ϵ-Greedy performance depends heavily on ϵ:

- $\epsilon = 0.05$: 2,592 clicks (nearly Thompson)
- $\epsilon = 0.1$: 2,551 clicks (good)
- $\epsilon = 0.01$: 2,503 clicks (too greedy)
- $\epsilon = 0.5$: 1,935 clicks (too much random exploration)

4.2 Why Thompson Sampling Wins

Thompson Sampling adapts as it learns. Early on, high uncertainty leads to exploration of all ads. As evidence accumulates, it focuses on the best ones. By round 7,000, it selects the optimal ad over 90% of the time.

ϵ-Greedy maintains constant exploration. Even late in learning, it randomly tries poor ads, wasting rounds.

4.3 Exploration Behavior

Thompson Sampling: Tries all ads early. By 10,000 rounds, it selected the optimal ad 9,486 times. Bad ads received only 24-55 selections total.

UCB: Explores more gradually. After 10,000 rounds, it selected the optimal ad 6,323 times.

ϵ-Greedy ($\epsilon = 0.1$): Selected optimal ad 9,057 times. But it explored all ads uniformly (90-122 selections each), regardless of quality.

4.4 Effect of ϵ

Performance peaks at $\epsilon \approx 0.05$. Too low ($\epsilon = 0.01$) misses good arms early. Too high ($\epsilon = 0.5$) wastes too many rounds on exploration. Thompson Sampling performs best with no tuning needed.

5 Conclusion

Thompson Sampling is the best strategy for this problem because:

1. Adapts exploration to learning progress
2. No parameter tuning required
3. Balances exploration-exploitation optimally

ϵ-Greedy works well if tuned ($\epsilon \approx 0.05$), but requires prior knowledge. UCB is theoretically sound but converges slower on this dataset.

Key insight: Smart, adaptive exploration beats fixed or random exploration.

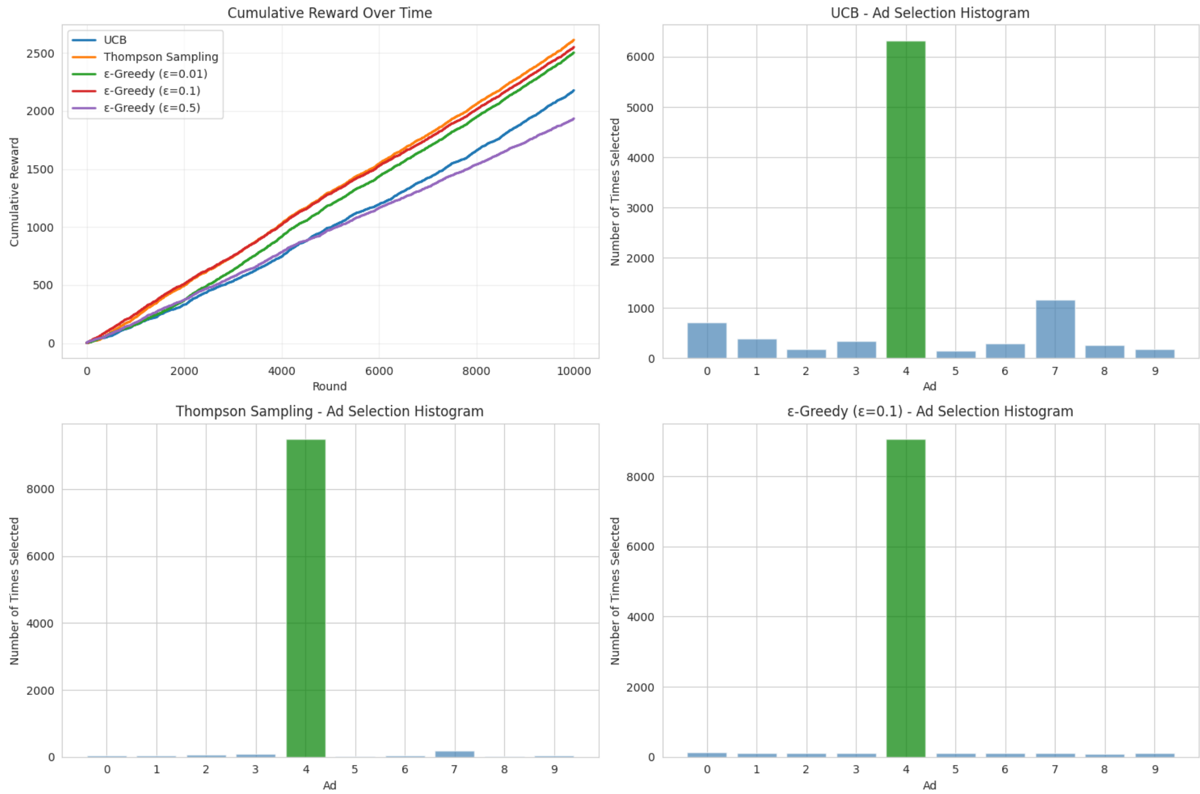

Figure 1: Cumulative reward curves and ad selection histograms

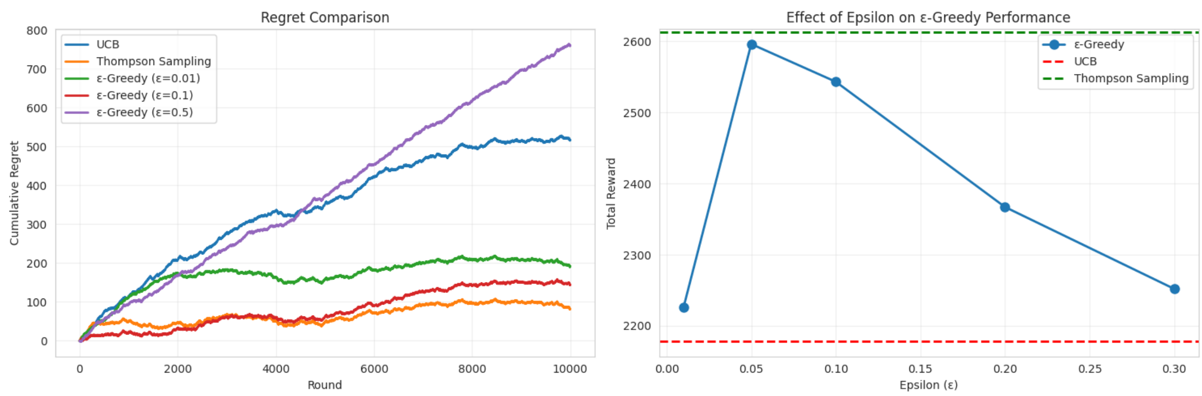

Figure 2: Regret curves and epsilon sensitivity

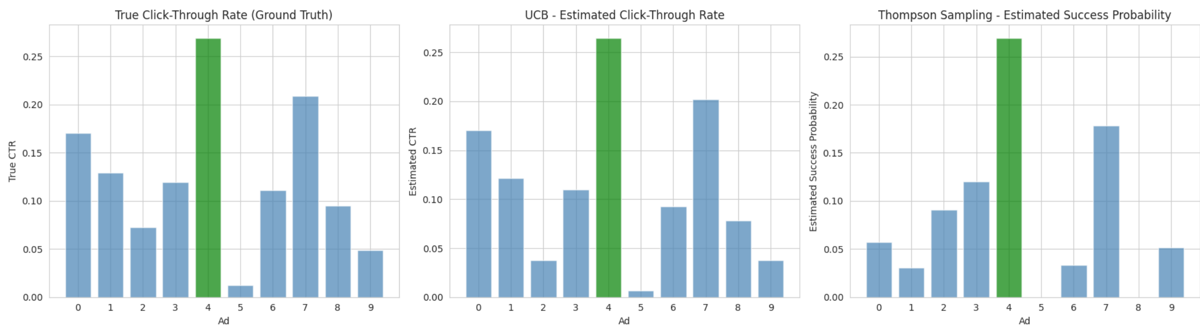

Figure 3: True vs estimated ad performance